Education City High School

2.3

Exponential Functions

Student Notes and Guided Practice

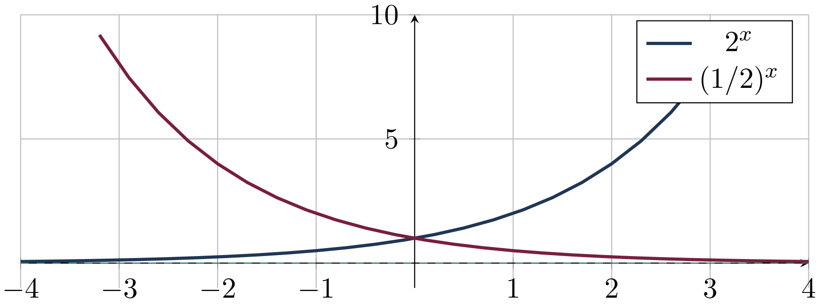
AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Analyze domain, range, asymptote, monotonicity, concavity, and end behavior of exponential functions.
- Interpret initial value and growth/decay factor.
- Analyze transformed exponential graphs.

Essential Question

How does the base of an exponential function control its behavior?



Concept Framework

Key Idea

For $b > 1$, b^x is increasing; for $0 < b < 1$, it is decreasing.

Key Idea

The parent exponential has domain all real numbers, range positive numbers, and horizontal asymptote $y = 0$.

Key Idea

A vertical shift moves the horizontal asymptote.

Key Idea

Nonzero exponential functions do not have x-intercepts unless shifted.

Formula Map

Formula and Structure

$$f(x) = ab^{x-h} + k$$

Formula and Structure

horizontal asymptote $y = k$

Formula and Structure

$$\frac{f(x+1)-k}{f(x)-k} = b$$

Worked Examples

Worked Example 1

Problem. Analyze $f(x) = 3(1/2)^{x-2} - 4$.
Solution. Domain all real; range $(-4, \infty)$; asymptote $y = -4$; decreasing and concave up.

Worked Example 2

Problem. Find the growth factor if an exponential doubles every 5 units.
Solution. One-unit factor is $2^{1/5}$.

Worked Example 3

Problem. Can $2^x + 1$ equal 0?
Solution. No. Since $2^x > 0$, $2^x + 1 > 1$.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Analyze domain, range, asymptote, monotonicity, and concavity of $3 \cdot 2^x$.

2. Analyze domain, range, asymptote, monotonicity, and concavity of $5(1/3)^x$.

3. Analyze domain, range, asymptote, monotonicity, and concavity of $-2^x + 4$.

4. Analyze domain, range, asymptote, monotonicity, and concavity of $2 \cdot 3^{x-1} - 5$.

5. Analyze domain, range, asymptote, monotonicity, and concavity of $-4(1/2)^{x+2} + 1$.

6. Find a formula $f(x) = ab^x$ through (0,6) and (2,24).

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is ____________________.
2. A restriction I must remember is ____________________.
3. One graphical feature I can justify algebraically is ____________________.

Video support: Flipped Math topic page | Original ECHS material informed by Pearson and Holt Precalculus concepts.